



**WESTMINSTER BUSINESS SCHOOL
OF
FINANCE AND ACCOUNTING**

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1. Introduction

This report explores a quarter-by-quarter data set of the US economy from 2004 to 2025, with the focus on understanding the linkage of the US GDP to various prominent macroeconomic factors. The work has been carried out using only Microsoft Excel, where the descriptive analysis, distribution checks, correlation and covariance, confidence intervals, hypothesis tests, regression modelling, diagnostics, time series tools, and forecasting have been applied. The variables in the regression are US GDP growth (dependent) against inflation (new), unemployment, interest rate, and debt to GDP (independent). The intent of the exercise is extended beyond the mere output computing; it is very much about the interpretation of the outputs and understanding what these imply for the predictability, risk, and investment decision-making.

2. Data Description

The data used for this analysis encompasses quarterly observations from 2004 to 2025, totalling 83. The US GDP growth is studied both in a log form as well as a raw form. Inflation was recalculated into a quarterly percentage change from its original form so that its change is comparable to other variables, which were all in the quarterly format. Most of the data were already at the quarterly frequency, and hence no further transformation was necessary. What distinguishes this sample is the concentration of extreme events, specifically the Global Financial Crisis, as well as the COVID-19 pandemic, both of which are significant since such events tend to determine volatility, distribution biases, and model reliability.

3. Question 1: Descriptive and Distributional Analysis of GDP Growth

3.1 Descriptive statistics and what they mean

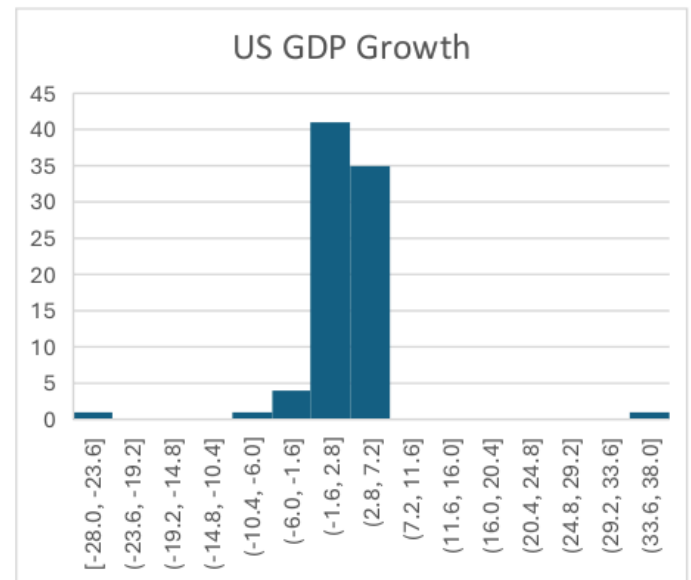
I have employed Excel Descriptive Statistics to get the descriptive parameters of the GDP growth in its raw form, and then for log GDP growth.

<i>GDP Level</i>			<i>GDP Log</i>	
Mean	2.272289		Mean	0.021014
Standard Error	0.60138		Standard Error	0.006031
Median	2.5		Median	0.024693
Mode	2.5		Mode	0.024693
Standard Deviation	5.478835		Standard Deviation	0.054944
Sample Variance	30.01764		Sample Variance	0.003019
Kurtosis	25.93252		Kurtosis	27.33786
Skewness	0.390328		Skewness	-1.73166
Range	62.9		Range	0.627868
Minimum	-28		Minimum	-0.3285
Maximum	34.9		Maximum	0.299364
Sum	188.6		Sum	1.744184
Count	83		Count	83

The average GDP growth stands at approximately 2.27, while the standard deviation is nearly 5.48; thus, quarterly growth is highly variable relative to its average. The range is incredibly vast, going from minus 28 to plus 34.9, which indicates that GDP growth is generally stable but is capable of rare, very large shocks. The kurtosis numbers are very high, which is consistent with fat tails, more frequent extreme outcomes than a normal distribution would suggest. This outcome is classed as a risk-relevant one since fat tails mean that the risk of crisis is not negligible.

3.2 Histogram and normality discussion

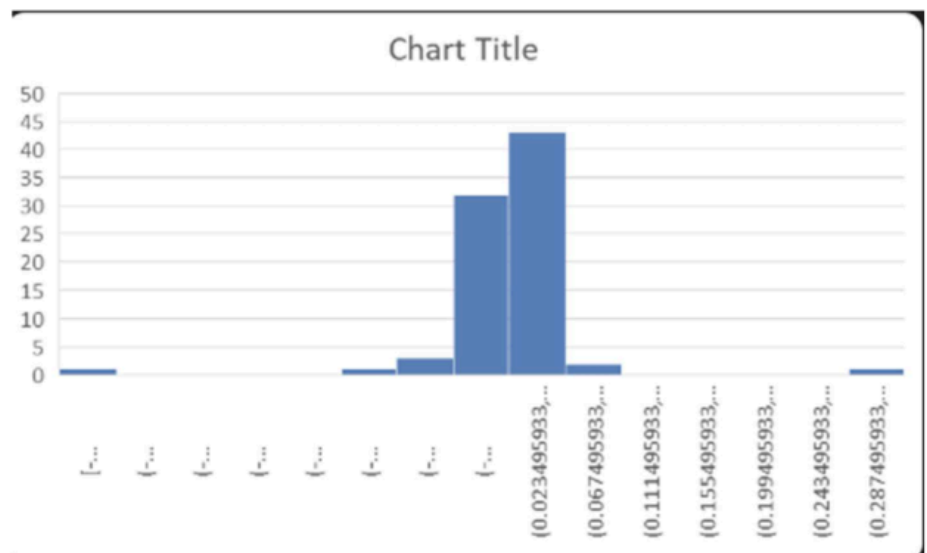
The histogram displays most observations grouped around moderate growth, but also with some outliers during the times of recession and the COVID shock. This shape does not look normal because it has extreme tails and visible asymmetry driven by downturn quarters.



3.3 Log transformation and why it matters

The log transformation squeezes the extreme values, and thus the distribution becomes more stable. From a visual point, the log histogram is more balanced, though it cannot be denied that there are still shocks. From the point of view of finance, the log form can be more appropriate for the model as extreme quarters do not dominate the estimation as much, thus the robustness for the statistical inference is

improved. Even so, the presence of macro shocks still hampers predictability.



4. Question 2 Correlation and Covariance Analysis

4.1 Correlation matrix interpretation

Corellation	US GDP Growth	New Inflation Rate	Unemployment Rate	Intrest Rate	DEPT to GDP
US GDP Growth	1				
New Inflation Rate	0.117167579	1			
Unemployment Rate	-0.174736102	-0.093936818	1		
Intrest Rate	0.025816388	0.076084637	-0.574425593	1	
DEPT to GDP	0.062082767	0.137837112	-0.057787343	-0.269584575	1

Your correlation results show the following key relationships.

GDP growth with the new inflation rate is about 0.117, a weak positive relationship. In real terms, inflation is not a significant short-term driver of GDP growth in this sample. It may be that they move in the same direction from time to time, but the relationship is too small to consider inflation as a reliable predictor on its own.

GDP growth with unemployment is about negative 0.175. The sign agrees with the economic theory, an increase in unemployment is usually associated with a decrease in growth, but the absolute value is still weak. This suggests that changes in the labour market do not convert into a clear linear GDP signal at the quarterly frequency, mainly because shocks and policy periods interfere with the relationship.

GDP growth with interest rate is about 0.026, almost zero. This means that within this dataset, interest rate levels are not linearly tracking GDP growth quarter by quarter. It does not mean that interest rates are not important in the economy. It indicates that the simple linear correlation is weak, probably because of policy lags and crisis periods.

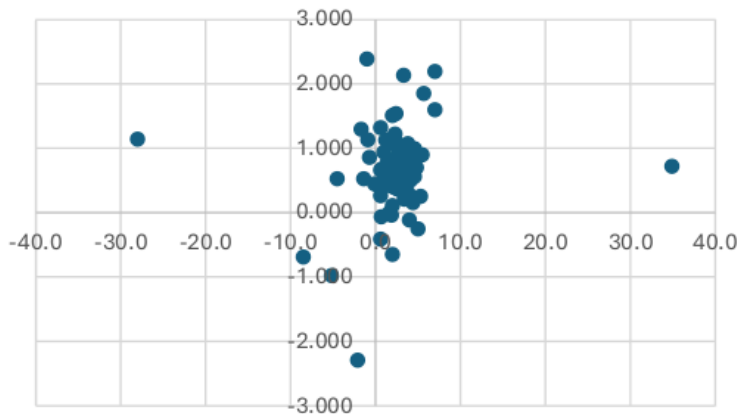
GDP growth with debt to GDP is about 0.062, again very weak. Debt to GDP acts as a slow-moving structural variable; thus, it does not show a strong short-term co-movement with GDP growth.

The strongest correlation in the matrix is that between unemployment and interest rate, approximately -0.574. This is significant. It mirrors the policy cycle during which interest rates tend to be lower at times of higher unemployment, thus being in line with the monetary easing during the downturns.

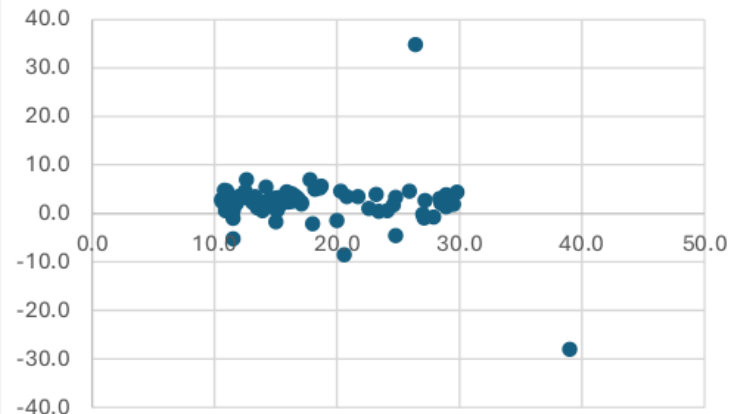
4.2 Scatter plots and what the plots should look like

For scatter plots, the proper configuration is:

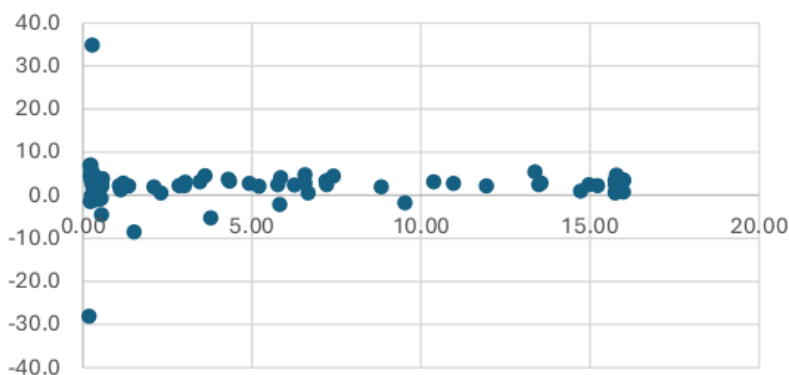
GDP Growth Vs Inflation rate



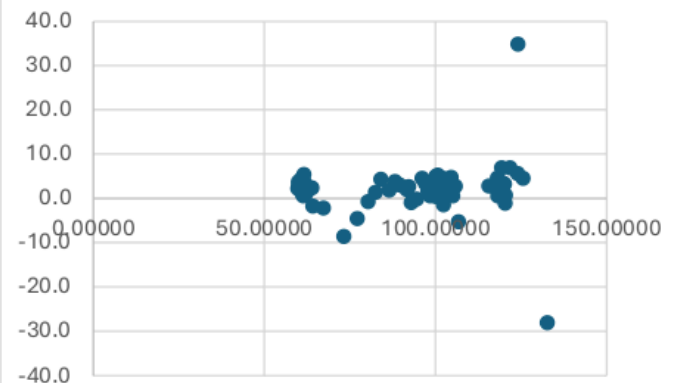
GDP Vs Unemployment Rate



GDP Vs Intrest Rate



GDP Growth Vs DEPT to GDP



Interpretation of your outputs:

The GDP inflation plot should resemble a cloud of points rather than a line, which corresponds to the weak correlation of 0.117. GDP vs. unemployment should reveal a slightly downward trend but still be scattered, thus indicating a weak negative correlation. The interest vs. unemployment plot should exhibit a more obvious downward trend, thus reflecting the moderately strong negative correlation of approximately -0.574.

Those plots are important for thinking about diversification. When correlations are close to zero, variables may be able to provide diversification benefits since a shock to one variable does not necessarily cause the other to move. However, the link between unemployment and interest shows that some macro variables become very tightly connected during periods of stress, thus diversification is reduced exactly when markets are under pressure.

4.3 Covariance matrix interpretation

<i>Covariance</i>	US GDP Growth	New Inflation Rate	Unemployment Rate	Intrest Rate	DEPT to GDP
US GDP Growth	29.6559791				
New Inflation Rate	0.418390679	0.429968678			
Unemployment Rate	-5.923800261	-0.383456309	38.75460009		
Intrest Rate	0.790748149	0.280609227	-20.11323733	31.63538575	
DEPT to GDP	7.072375205	1.890700195	-7.52544959	-31.71901553	437.5988629

Your covariance findings are in line with the direction of the relationships, but they depend on the scale.

The covariance between GDP growth and unemployment is close to -5.924, which is a good example of them moving in opposite directions on average. The covariance between Unemployment and Interest rate is close to -20.113, which is the biggest in absolute terms, and therefore, it is in line with the strong negative correlation.

Debt to GDP carries a very large variance of approximately 437.599, which shows that it fluctuates on a much larger numerical scale than the other variables. This is precisely why correlation is generally more useful than covariance when comparing the strength of relationships across different units.

5. Question 3: 95 Per cent Confidence Intervals

5.1 Description

Variable	Mean	Standard Error	Margin Of error	lower Bound	Upper Bound	Sample Size	Standard Deviation
US GDP Growth	2.3	0.601380314	1.178705415	1.1	3.5	83	5.478835404
New Inflation Rate	0.640	0.072412157	0.141927828	0.5	0.8		0.659706146
Unemployment Rate	17.5	0.687471508	1.347444156	16.1	18.8		6.263163511
Intrest Rate	4.90	0.621125903	1.21740677	3.7	6.1		5.658726282
DEPT to GDP	94.41373	2.310102054	4.527800025	89.9	98.9		21.04603132

Confidence interval table showing mean, standard error, margin of error, lower bound, and upper bound.

Inflation is the series with the most stable mean estimation in your sample, as it has the narrowest interval. GDP growth is a highly volatile series and, therefore, it has a wide interval with extreme shocks included. Debt-to-GDP also has a wide interval because it changes structurally over time.

Financial relevance: Wider confidence intervals imply that there is more uncertainty about the true average level. If an investor or analyst uses expected growth as a basis for making decisions, a wide interval serves as a warning that the mean is not tightly estimated and that risk buffers are necessary. The fact that inflation is tightly estimated is good because inflation assumptions are used in discount rates, real returns, and purchasing power, so stability facilitates planning.

6. Question 4 Hypothesis Testing

6.1 Single-sample t-test

Hypothesised Mean	2	As the absolute value of the t statistic is less than the critical value, the null hypothesis cannot be rejected at the 5% level.
T - Static	-1.02779	
Critical Value	1.989319	Statistically, your sample does not provide substantial evidence that long-run average quarterly GDP growth is different from 2.
Mean GDP Growth	2.3	
STD	5.48	Financially, this result is in line with the idea that long-run growth is still anchored around a stable benchmark, although there is a lot of short-run volatility.
Sample Size	83	
Standard Error	0.601508	
Hypothesised Mean Difference	0	
Alpha	0.05	

6.2 Two-sample t-test between two periods

t-Test: Two-Sample Assuming Unequal Variances		
	Variable 1	Variable 2
Mean	1.72	2.447619
Variance	9.195368	36.75318
Observations	20	63
Hypothesized Mean Difference	0	
df	65	
t Stat	-0.71241	
P(T<=t) one-tail	0.23938	
t Critical one-tail	1.668636	
P(T<=t) two-tail	0.47876	
t Critical two-tail	1.997138	

The P-value is much greater than 0.05; therefore, you fail to reject the null hypothesis that the means of the two periods are equal. There are differences in averages in the sample, but they are not statistically significant when compared to the level of volatility.

Financial meaning: It is a strong visual proof that GDP growth regimes might appear different, but when volatility is high, average differences become hard to confirm statistically. For investors, it is a signal not to overreact to short windows of growth performance.

7. Question 5 Simple Linear Regression

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.117167579							
R Square	0.013728242							
Adjusted R Square	0.001552047							
Standard Error	5.474582047							
Observations	83							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	1	33.79132899	33.79133	1.127466	0.291472209			
Residual	81	2427.654936	29.97105					
Total	82	2461.446265						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	1.649282845	0.839852965	1.963776	0.052987	-0.021761114	3.320326803	-0.021761114	3.320326803
New Inflation Rate	0.973072457	0.916417793	1.061822	0.291472	-0.850311492	2.796456406	-0.850311492	2.796456406

R-squared is very close to zero, which implies that inflation explains less than 1.4 per cent of the change in GDP growth. The slope is positive but statistically insignificant; thus, it is not possible to assert a linear impact of inflation on GDP growth with a certain level of confidence in this sample.

Financial implication: One of the main takeaways is that inflation alone is not a good predictor of GDP growth. Even if the sign of the coefficient is as expected, the uncertainty is too large. Therefore, an investor who would use inflation as the only signal for timing growth would be exposed to a high model risk.

8. Question 6 Multiple Regression

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.222708736							
R Square	0.049599181							
Adjusted R Square	0.000860678							
Standard Error	5.476477141							
Observations	83							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	4	122.0857191	30.52142979	1.01765909	0.403533746			
Residual	78	2339.360546	29.99180187					
Total	82	2461.446265						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	5.665468262	4.547500399	1.245842279	0.216551496	-3.387909578	14.7188461	-3.387909578	14.7188461
New Inflation Rate	0.860555201	0.932313671	0.923031838	0.358837132	-0.995538738	2.71664914	-0.995538738	2.71664914
Unemployment Rate	-0.200829746	0.12252621	-1.639075809	0.105225057	-0.444760692	0.043101199	-0.444760692	0.043101199
Intrest Rate	-0.109247602	0.141057784	-0.774488289	0.440983483	-0.390072077	0.171576874	-0.390072077	0.171576874
DEPT to GDP	0.001071216	0.031365268	0.03415293	0.97284245	-0.061372236	0.063514667	-0.061372236	0.063514667

Interpretation: The model increases R-squared compared to the simple regression only up to about 5 per cent. The adjusted R-squared is practically zero, which means that accounting for complexity, the new variables do not have a strong explanatory power. Also, none of the variables are statistically significant at the 5 per cent level.

Financial meaning: Short-run GDP growth cannot be easily explained by a small number of macro variables using a linear model, especially when there are crisis observations. Even though the signs are mostly in line with the economic theory, the weak statistical significance of the model implies that it should not be used as a standalone forecasting tool.

9. Question 7 Diagnostics

9.1 Autocorrelation

Durbin-Watson statistic 2.427939

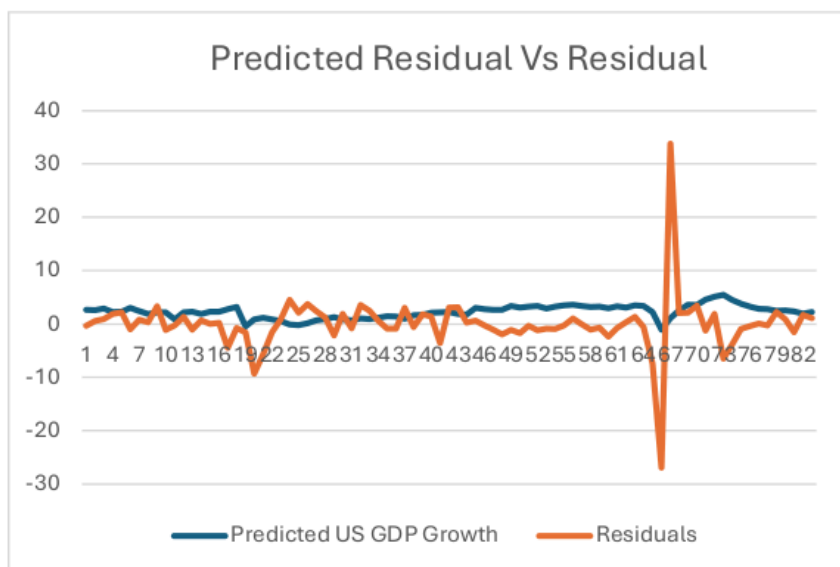
The Durbin Watson statistic is 2.428. Values close to 2 mean that there is no serious first-order autocorrelation. Thus, the residuals are not strongly dependent on time, and hence, the inference of the coefficients is not greatly affected by serial correlation.

9.2 Multicollinearity

Corellation	US GDP Growth	New Inflation Rate	Unemployment Rate	Intrest Rate	DEPT to GDP
US GDP Growth	1				
New Inflation Rate	0.117167579	1			
Unemployment Rate	-0.174736102	-0.093936818	1		
Intrest Rate	0.025816388	0.076084637	-0.574425593	1	
DEPT to GDP	0.062082767	0.137837112	-0.057787343	-0.269584575	1

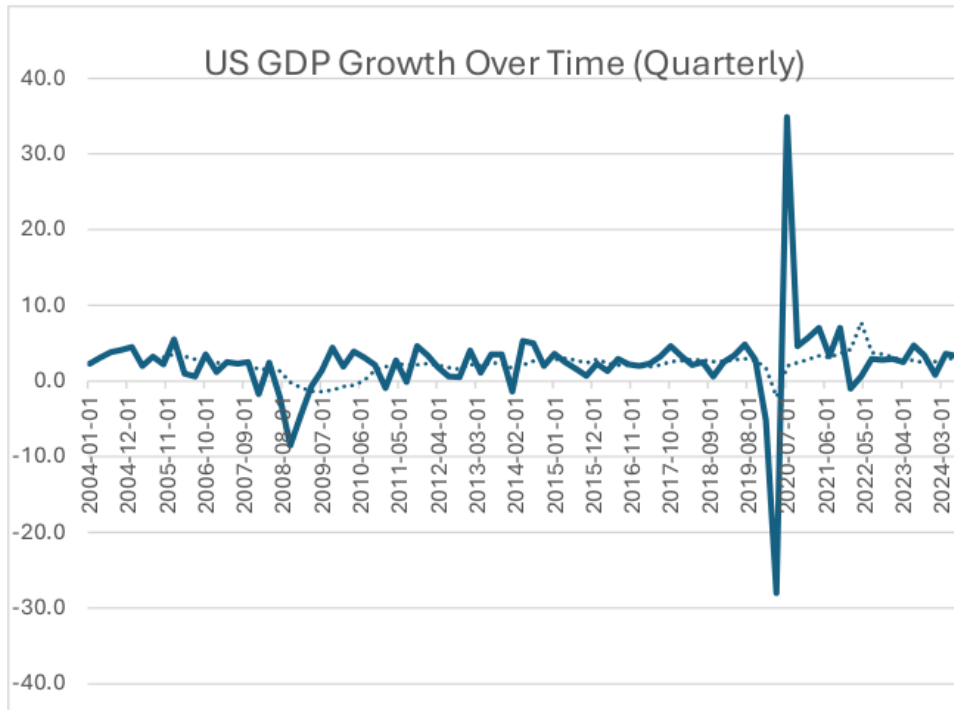
The independent variables are not strongly correlated with each other; however, the relationship between unemployment and interest rate is moderately strong. None of the numbers are close to +1 or -1, so multicollinearity is not a big problem. In other words, the instability of the coefficients is more probably the result of a weak signal and high volatility rather than the presence of redundant predictors.

9.3 Linearity and homoscedasticity



Based on scatter style relationships and the behaviour of the residuals, there is no obvious evidence of strong non-linear patterns. The variance of residuals appears to be quite stable; however, there are some large residuals at the crisis points. These large residuals reflect real macro shocks, not necessarily a modelling mistake. The main issue is still low explanatory power, not violated assumptions.

10. Question 8 Time Series Tools and Moving Average



Interpretation: The series is a typical business cycle, one with two very clear periods of shocks: 2008 to 2009 and 2020. The moving average gets rid of the noise and reveals the underlying long-run trend. Among the most important implications is the fact that extreme shocks account for most of the short-run volatility, but the moving average shows that after the shocks, the economy tends to go back to a stable growth path.

Investment implication: Short-term GDP signals are noisy and risky if used for timing. MAs enable investors to refrain from overreacting to one quarter's results and instead focus on the broader regime direction.

11. Question 9 Forecasting and Error Metrics

2024-10-01	4.5	3.3
2025-01-01	3.6	3.3
2025-04-01	3.8	3.4
2025-07-01	4.2	3.4

MAE	0.7	MSE	0.5825
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Your forecast evaluation is based on four quarters where both actual and forecast values are available: MAE is equal to 0.7, MSE is equal to 0.5825

Interpretation: MAE of 0.7 means that the forecast deviates from the actual value by 0.7 percentage points on average during the evaluated period. The MSE is also quite low, which indicates that there is no significant error explosion.

Limitations: This performance measure is only four observations; thus, it cannot be taken as strong evidence of long-term forecasting quality. Moreover, simple forecasting methods usually smooth the series and cannot predict shocks such as 2020.

Financial relevance: The forecast can be regarded as a short-horizon baseline in quiet periods, but it should not be relied upon for stressful scenarios. Investors should treat it as a reference line rather than a decision rule.

12. Question 10 Recommendations and Reflection

Throughout the analysis, GDP growth has been portrayed as a stable long-term average most of the time, but with very high risks from quarter to quarter, especially for crisis outliers. The distribution analysis reveals that fat tails are present, which means that rare events are more likely than what the normal assumptions indicate. This is quite significant for investors, as it gives more weight to the importance of stress testing and diversification as a strategy rather than solely depending on average-based expectations.

The correlation findings reveal that GDP growth has only weak linear associations with inflation, unemployment, interest rates, and debt-to-GDP. The dataset depicts the strongest relationship between unemployment and interest rates, which is in line with policy responses to downturns. From a portfolio point of view, this indicates that some macro variables get more connected during the stress periods, thus the diversification is reduced precisely at the time it is needed most.

The confidence intervals depict inflation as the most predictable mean estimate, whereas GDP growth and debt-related measures have wider uncertainty bands. This points to the aspects of measurement and estimation risk: historical averages should not be seen as precise guarantees.

The hypothesis tests endorse the view that long-run growth is not statistically different from a 2 per cent benchmark and that mean growth across the two periods tested is not statistically different, given the presence of volatility. This is an argument for the necessity of long-horizon thinking.

The regression results agree with the idea that the selected macro variables have weak short-run predictive power for GDP growth. According to diagnostics, the multiple regression model is statistically usable; however, its low R-squared indicates that its explanatory strength is very limited. This suggests that GDP growth depends on wider structural and shock-driven factors apart from just a few indicators.

Time series instruments and moving averages reveal that although growth is noisy, the underlying trends become clearer when the short-run volatility is smoothed. The forecasting results indicate a reasonable short-horizon accuracy in the evaluation window; however, the evaluation sample is small, and the method is not intended for crisis prediction.

In brief, the recommendation is to: (1) refrain from making very precise quarter forecasting of GDP growth and instead use these data along with macro indicators for regime context; (2) combine the quantitative outputs with scenario analysis, stress testing, and qualitative judgment. Future work can be practically improved by testing models that handle structural breaks, have larger predictor sets, or use alternative forecasting tools which can adapt to shocks.

References

- Blanchard, O. (2017) *Macroeconomics*. 7th edn. Harlow: Pearson Education.
- Box, G.E.P., Jenkins, G.M., Reinsel, G.C. and Ljung, G.M. (2015) *Time Series Analysis: Forecasting and Control*. 5th edn. Hoboken: Wiley.
- Brooks, C. (2019) *Introductory Econometrics for Finance*. 4th edn. Cambridge: Cambridge University Press.
- Gujarati, D.N. and Porter, D.C. (2009) *Basic Econometrics*. 5th edn. New York: McGraw-Hill.
- Hamilton, J.D. (1994) *Time Series Analysis*. Princeton: Princeton University Press.
- Mankiw, N.G. (2021) *Principles of Macroeconomics*. 9th edn. Boston: Cengage Learning.
- Stock, J.H. and Watson, M.W. (2019) *Introduction to Econometrics*. 4th edn. Harlow: Pearson Education.
- Taylor, J.B. (1993) 'Discretion versus policy rules in practice', *Carnegie-Rochester Conference Series on Public Policy*, 39, pp. 195–214.
- Wooldridge, J.M. (2020) *Econometric Analysis of Cross Section and Panel Data*. 2nd edn. Cambridge, MA: MIT Press.
- Federal Reserve Bank of St. Louis (2024) *FRED Economic Data*. Available at: <https://fred.stlouisfed.org> (Accessed: 2025).
- U.S. Bureau of Economic Analysis (2024) *National Income and Product Accounts*. Available at: <https://www.bea.gov> (Accessed: 2025).